FleetCentric research plan

Route optimization and rerouting for FAST NUCES Karachi shuttles · Overview as of 3 Nov 2026

In one line: can a free-tier server build better all-stop shuttle routes than the current hand-made ones, and reroute a bus sensibly when a road is blocked, within tight CPU and query limits?

Scope

Phase	Question
FYP-1	**Optimization.** How close can an optimizer get to its best possible routes under free-tier limits (512 MB, 0.1 CPU, fixed time and query budgets), and how do those routes compare with the current routes?
FYP-1	**Rerouting.** When a road is blocked, which road and which stops (keep, skip or merge) give the lowest total student delay, computed within the same limits?
FYP-2	**Context.** How much real incidents slow shuttle routes, recorded on an Uber H3 hexagon grid.
FYP-2	**Prediction.** Whether that context predicts travel times better than the timetable and the router.

Research lives in the `research/` folder only. It does not change the website, backend, settings or database, and is built into the product only once it is proven.

Sub-questions for optimization

- **Time budget:** how route quality changes from 10 s to 600 s of search under the CPU cap, and where extra time stops helping.
- **Query budget:** how few real travel-time queries (each stop's K nearest stops, the rest estimated) still give good routes.
- **Traffic-aware matrix** (later): whether routes planned on rush-hour times beat routes planned on free-flow times.

Data

	Jadoon	Nadeem	Total
Stops	171	143	311 (3 shared)
Current routes	27	25	52
Buses / seats	27 / 1,220	26 / 1,195	53 / 2,415
Students (synthetic)	906	900	1,806

- Real Fall 2026 route plan and stop locations; 11 stops without locations placed from their route neighbours.
- Students are synthetic (no access to real data yet): each current bus filled to 65 to 90%, fixed random seed, so current routes are feasible and both plans serve the same students.

- Each contractor plans only its own stops with its own buses.

Method

- **Road network:** self-hosted OSRM (free, open source) on a Karachi OpenStreetMap extract, run in Docker on WSL. 410k nodes; customizing after a road change takes 4.5 s, which rerouting will rely on.
- **Travel times:** one OSRM call per contractor gives full matrices (172×172 and 144×144). All pairs reachable. Times are free-flow (no traffic).
- **Optimizer:** Google OR-Tools. Rules: every stop visited, bus capacity, arrive at campus by 08:00, no ride over 90 min, stop time 60 s + 5 s per student. Goal: lowest total student ride time.
- **Fair comparison:** current routes are scored with the same function on the same matrix.
- **Record:** every decision and run is logged, dated, in `research/LOGBOOK.md`, the source of truth for the paper.

Progress so far

Run	What	Result
D1	Dataset built	Reproducible, identical on two machines
N1	Road network built	Partition 6.5 s / 214 MB, customize 4.5 s / 291 MB
M1	Travel-time matrices	2 calls, 0 unreachable pairs, 9 to 40 min to campus
O0	Optimizer smoke test (20 s, not limited)	Total student ride time down 21.5% (Jadoon) and 19.7% (Nadeem); longest ride 78.7 to 56.1 min (Jadoon); peak memory under 65 MB

O0 shows the method works; it is not a reported result. Savings rest on free-flow times and synthetic students.

Next steps

#	Step	Output
1	Time-budget experiment: 10/60/300/600 s, limited (512 MB, 0.1 CPU) vs unlimited, 3 repeats, against a 30-min reference	Quality-vs-time chart, memory and time table
2	Query-budget experiment: K nearest stops queried, rest estimated	Quality vs number of queries
3	Rerouting: block roads on route segments, compare keep-order detour, visit-all reorder and keep/skip/merge	Student delay, students walking, decision time
4	Write-up from the logbook	FYP-1 research report

Proposed thresholds, to be fixed before step 1 runs: limited setup within 5% of the reference at 60 s; peak memory under 400 MB.

Known limits

- Free-flow travel times; real morning traffic will differ.
- Synthetic students; real counts may change the savings.
- Current routes may follow rules the model cannot see (narrow roads, driver areas, student requests).

- No bus GPS yet, so no real trip times to validate against.